

MyCCAngel: Case Study

Designing an AI-Powered Healthcare Community Platform for Individuals Living with Inflammatory Bowel Disease (IBD)

Executive Summary

Project Overview

Role: Founding Product Lead

Duration: 2024 – Present (Development currently paused)

Team Size: Up to 8 team members (Frontend Developers, Backend Developers, UX/UI Designer, Data & Research Contributors)

Project Type: Healthcare Technology · Artificial Intelligence · Community Platform · Startup

Project Background

MyCCAngel was conceived as an AI-powered digital platform designed specifically for individuals living with Inflammatory Bowel Disease (IBD). The platform aimed to create a trusted ecosystem where verified patients could connect with one another, access reliable healthcare resources, receive personalized recommendations, and find practical support for managing the daily challenges associated with IBD.

Rather than functioning as another general social network, MyCCAngel was designed around the unique needs of IBD patients by combining artificial intelligence, community engagement, educational resources, and healthcare support into a single platform.

As Founding Product Lead, I was involved from the earliest stages of product discovery through requirements gathering, feature planning, UX collaboration, AI feature development, and coordination with designers and developers during implementation.

Problem Statement

Living with IBD affects far more than physical health. Patients often experience disruptions to work, education, travel, relationships, mental wellbeing, and everyday activities.

During research, several recurring problems became apparent. Patients struggled to:

- Find trustworthy online communities consisting of verified IBD patients
- Access reliable healthcare information in one centralized location
- Discover specialists, dietitians, organizations, and educational resources relevant to their condition
- Receive personalized guidance based on their symptoms and disease characteristics
- Manage everyday challenges such as locating accessible toilets while traveling or understanding dietary options

Although many organizations provide pieces of this support, there was no single platform bringing these resources together while also fostering meaningful community engagement. This gap became the foundation for MyCCAngel.

Product Vision

The vision for MyCCAngel was to become a trusted digital ecosystem supporting individuals throughout their IBD journey. The platform was designed around five strategic pillars:

- Community Engagement
- AI-Powered Personalization

- Educational Resources
- Healthcare Support Services
- Daily-Life Assistance

Rather than building features for their own sake, every planned capability was intended to support one or more of these pillars.

Product Strategy

Rather than attempting to build a fully featured healthcare ecosystem immediately, the strategy focused on delivering a Minimum Viable Product (MVP) capable of demonstrating the platform’s value. The MVP was designed to achieve four objectives:

- Demonstrate the complete product vision
- Validate user interest
- Support discussions with healthcare organizations and clinicians
- Provide a foundation for future partnerships and funding opportunities

Several features — including doctor directories, partner organizations, community groups, and events — were intentionally postponed until after the platform attracted users and strategic partners. This phased approach reduced development complexity while maximizing learning opportunities.

My Contributions

As Founding Product Lead, I worked across product strategy, design, AI development, and project coordination.

Area	What I Did
Product Strategy	Defined product vision, identified target users, conducted competitive analysis, prioritized features, created roadmaps, defined MVP strategy
Product Discovery	Designed patient onboarding process and questionnaire, defined user journeys, identified personalization opportunities, gathered patient and stakeholder feedback
Product Design	Defined platform functionality and user flows, initiated majority of platform features, collaborated with UX/UI designer in Figma, reviewed and refined interface designs
Product Development	Collaborated daily with frontend and backend developers, prioritized work, clarified functional requirements, coordinated implementation across contributors
AI & Machine Learning	Designed and built the SVM medical report classifier (~91% accuracy), built the recommendation engine, created and labeled 200-report dataset
AI Chatbot	Defined requirements and designed the full UI/UX; development team built the NLP/ML pipeline (functional, IBD training pending)

User Research & Discovery

Understanding patient needs was essential before defining the platform. A structured onboarding experience was created to both verify users and collect the information required for personalization.

Patient Verification

Users uploaded a medical report during registration. The report was analyzed through an AI verification workflow before the account could be approved.

Patient Profile

The onboarding survey collected:

- Demographics
- Disease Type
- Disease Duration
- Disease Status
- Symptom Severity
- Toilet Frequency
- Daily Challenges
- Areas requiring support

Unlike traditional registration forms, each question served a functional purpose within the product. The collected information supported user verification, personalized recommendations, future analytics, and community matching.

MVP Strategy

The original vision included many features. However, resource limitations required careful prioritization. The MVP focused on building a platform that could realistically demonstrate the product's value.

Included within the MVP

- Registration & Authentication
- Medical Report Verification
- User Profiles
- Personalized Onboarding
- Recommendation Engine
- Direct Messaging
- Resource Library & Educational Articles
- AI Chatbot (NLP/ML pipeline built by the team; UI/UX designed by me; IBD-specific training was the next step before the project paused)
- Toilet Locator

Features intentionally postponed

- Doctor Directory
- Dietitian Directory
- Partner Organizations
- Events
- Community Groups
- Advanced AI Personalization
- Restaurant Recommendations

This phased approach allowed development resources to remain focused while creating a compelling demonstration platform.

Product Decisions

Several important product decisions shaped the platform.

Trust Before Growth

Rather than maximizing registrations, user verification became mandatory. A trusted community was considered more valuable than rapid user growth.

Solve the Cold-Start Problem

Since the platform initially had very few users, the recommendation engine relied primarily on onboarding data instead of behavioral data. The architecture was designed to evolve as user activity increased.

Build Partnerships After the MVP

Healthcare directories were intentionally postponed until a working platform existed that could be demonstrated to doctors, dietitians, and organizations.

Prioritize Everyday Challenges

Rather than focusing only on medical information, the platform emphasized practical daily-life support, including toilet location assistance and personalized resources.

Artificial Intelligence

Medical Report Verification

One of the platform’s most innovative features was an AI-powered verification system that classified uploaded medical reports before granting community access.

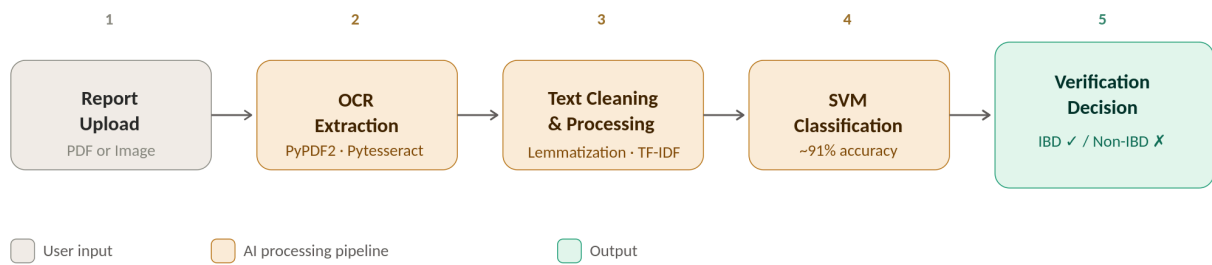


Figure 1: Medical report verification pipeline

The model was trained and evaluated on a manually collected and labeled dataset of 200 medical reports (held-out test split). A Support Vector Machine (SVM) classifier achieved approximately 91% classification accuracy. This significantly reduced manual review while increasing trust within the community.

Recommendation Engine

The recommendation engine addressed one of the biggest challenges facing new platforms: helping users quickly discover relevant people and content from the moment they join.

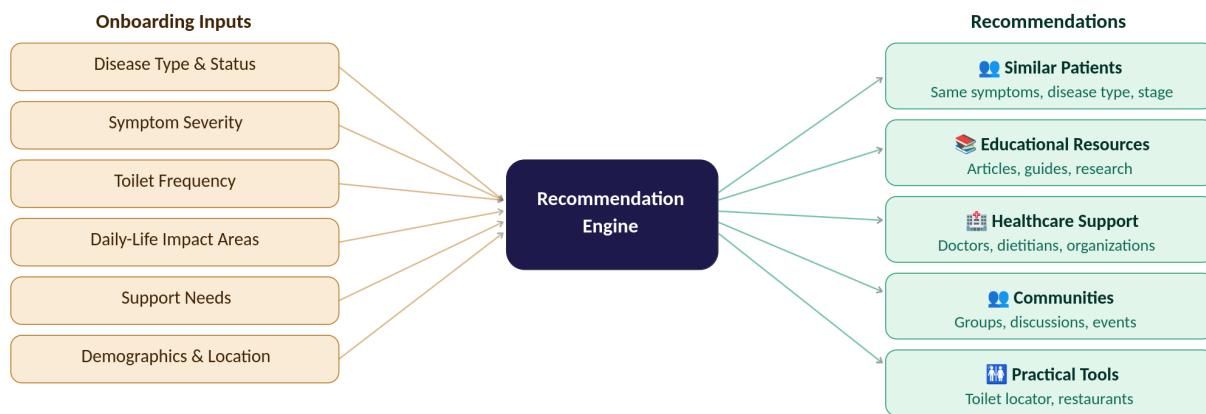


Figure 2: Recommendation engine — onboarding inputs mapped to personalized outputs

The system was intentionally designed so future versions could incorporate user behavior, interactions, and engagement history as the platform grew.

AI Healthcare Assistant

I defined the requirements and designed the full UI/UX for an AI-powered healthcare assistant. The development team built the underlying NLP/ML conversational pipeline, which was functional and able to hold a conversation by the time development paused.

The remaining step before launch was training the pipeline on a curated IBD knowledge base — trusted medical literature, partner content, and FAQ material — so that responses would be accurate and on-topic.

The assistant was designed to:

- Explain healthcare concepts in plain language
- Recommend relevant platform resources
- Guide users through available features

To maintain patient safety, the assistant was never intended to diagnose conditions or recommend treatments.

Product Development Process

Development followed an iterative product lifecycle.

Discovery

Problem identification, user research, competitive analysis, feature discovery.

Definition

Requirements, user journeys, feature prioritization, technical planning.

Design

Wireframes, UX design, UI design, interactive Figma prototypes.

Development

Frontend development, backend development, AI development, integration, and testing.

Current Status

Significant progress was completed before development paused. Completed work included:

- Product strategy and roadmap
- Requirements gathering

- UX/UI design and interactive Figma prototypes
- Medical report verification model (SVM, ~91% accuracy)
- Recommendation engine
- AI chatbot UI/UX design; NLP/ML pipeline built by the team (IBD training pending)
- Significant frontend implementation — all core pages designed and built
- Backend development underway on the signup/verification chain (paused mid-build)

Development paused after several contributors left the project, reducing the team from approximately eight members to four. The team is currently evaluating future development options.

Challenges

Resource Constraints

Building an ambitious healthcare platform with a small startup team required constant prioritization.

Feature Scope

Balancing long-term vision against short-term deliverables required careful product decisions.

Healthcare Complexity

Healthcare products require greater emphasis on trust, safety, privacy, and information quality.

Team Continuity

Changes in volunteer availability ultimately delayed platform completion.

Lessons Learned

- Understanding user problems is more valuable than simply adding features
- AI delivers the greatest value when integrated into a thoughtfully designed user experience rather than implemented as standalone functionality
- Early-stage startups benefit from validating smaller MVPs before expanding feature scope
- Strong communication between product, design, and engineering teams is essential for maintaining alignment and momentum

Reflection

MyCCAngel provided practical experience leading the development of an AI-enabled healthcare platform from concept through implementation. The project strengthened my abilities in product strategy, user discovery, AI product design, cross-functional collaboration, and startup execution.

Although development paused before launch, the experience demonstrated the importance of balancing ambitious product visions with available resources while continually prioritising user value.